**13** **(a)** Use triangle ABE in **Fig. C2** to show that $\arctan x + \arctan\left(\frac{1}{x}\right) = \frac{\pi}{2}$, as given in line 29. **[1]**

**(b)** Sketch the graph of $y = \arctan x$. **[1]**

**(c)** What property of the arctan function ensures that $y > \frac{1}{x} \Rightarrow \arctan y > \arctan\left(\frac{1}{x}\right)$, as given in line 30? **[1]**

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| **13(a)** | |
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| **13(b)** | |
| **13(c)** | |
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